Adaptive Neighborhood Selection and Metric Standardization for Local Dimension Estimation on Hydrodynamic Attractors

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Abstract

Determining the local intrinsic dimension of non-equilibrium solutions in relativistic hydrodynamics via Local Principal Component Analysis (LPCA) and the Participation Ratio (PR) provides a quantitative diagnostic of attractor formation. However, the choice of the neighborhood size K and the phase-space metric is fundamentally coupled to the aspect ratio of the sampling domain and the multi-scale relaxation dynamics of non-hydrodynamic modes. In this work, we present a complete mathematical derivation of the optimal neighborhood size $K_{\text{opt}}(N, D) \approx \text{clamp}(\text{round}(N^{0.45}\sqrt{D}), 12D, \lfloor N/10 \rfloor)$, derived by formalizing the trade-off between Marchenko–Pastur spectral noise in random matrix theory, tangent-space approximation bias, and attractor curvature. Furthermore, we demonstrate that coordinate standardization (Z-score) acts as an empirical diagonal Mahalanobis metric, rendering the K-nearest neighbor search invariant under rectangular domain disparities (such as $\Delta T \times \Delta \mathcal{A}$). Applied to conformal Müller–Israel–Stewart (MIS) and Heller–Janik–Spaliński–Witaszczyk (HJSW) models, this framework resolves the three-stage transition into hydrodynamization: early-time expansion, transient bimodal collapse, and asymptotic one-dimensional attractor evolution.

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1 Introduction and Physical Context

Relativistic heavy-ion collisions produce strongly coupled Quark-Gluon Plasma (QGP), whose early expansion is characterized by large gradients and substantial pressure anisotropies $\mathcal{A} \equiv (\mathcal{P}_L - \mathcal{P}_T)/\mathcal{E}$. Despite being far from local thermodynamic equilibrium, macroscopic hydrodynamic behavior sets in on timescales $\tau \lesssim 1\,\mathrm{fm}/c$. This phenomenon, termed *hydrodynamization*, is explained by the existence of hydrodynamic attractors: universal low-dimensional manifolds in phase space onto which vast ensembles of initial conditions rapidly collapse [3, 6, 7, 5].

To characterize this phase-space compression rigorously, dimensional reduction and intrinsic dimension estimation techniques such as Local Principal Component Analysis (LPCA) have recently been applied [9, 2]. Specifically, the local Participation Ratio (PR) dimension,

$$d_{\mathrm{PR}}(\mathbf{x}_i) = \frac{(\mathrm{Tr}\,\mathbf{C}_i)^2}{\mathrm{Tr}\left(\mathbf{C}_i^2\right)} = \frac{\left(\sum_{j=1}^{D} \lambda_j^{(i)}\right)^2}{\sum_{j=1}^{D} \left(\lambda_j^{(i)}\right)^2}, \tag{1}$$

evaluates the effective dimension spanned by the K nearest neighbors of point $\mathbf{x}_i$ without requiring artificial spectral cutoffs or singular-value thresholding. Here, $\mathbf{C}_i \in \mathbb{R}^{D\times D}$ is the empirical covariance tensor computed over the local neighborhood $\mathcal{N}_K(\mathbf{x}_i)$, and $\lambda_j^{(i)}$ are its eigenvalues.

Crucially, the hydrodynamic solutions approach the attractor across three physically distinct stages:

1. **Stage 1: Early-time expansion** ($\tau \sim 0.20$–$0.35\,\mathrm{fm}/c$)**:** The system retains complete memory of its initial conditions. Trajectories fill a full D-dimensional volume in phase space ($D = 2$ for conformal MIS, $D = 3$ for HJSW).

2. **Stage 2: Non-hydrodynamic mode damping** ($\tau \sim 0.40$–$0.70\,\mathrm{fm}/c$)**:** Non-hydrodynamic transients decay exponentially as $\delta\mathcal{A}(\tau) \propto \exp\left(-\int \frac{d\tau}{\tau_\pi}\right)$. Solutions collapse toward the universal attractor, generating a dense core surrounded by transient wings.

3. **Stage 3: Asymptotic attractor evolution** ($\tau \gtrsim 0.80\,\mathrm{fm}/c$)**:** Memory of initial anisotropy is entirely erased, and all solutions evolve along a universal one-dimensional curve ($d \approx 1.0$).

Despite this intuitive physical progression, raw histograms of d_{PR} frequently appear to remain 2-dimensional even at intermediate times ($\tau \approx 0.55\,\mathrm{fm}/c$) if the neighborhood parameter K is chosen naively. As shown below, this apparent paradox stems from the multi-scale coupling between K, the sample size N, and the aspect ratio of the phase-space coordinates.

2 Domain Aspect Ratio and Metric Invariance via Standardization

A central difficulty in phase-space manifold analysis is that the physical features have different dimensions and disparate numerical spans. For example, in conformal MIS, the initial anisotropy spans $\Delta\mathcal{A} \sim 4$, whereas temperature spans $\Delta T \sim 5\,\mathrm{fm}^{-1}$ (or $\Delta T \sim 300\,\mathrm{MeV}$ in particle physics units).

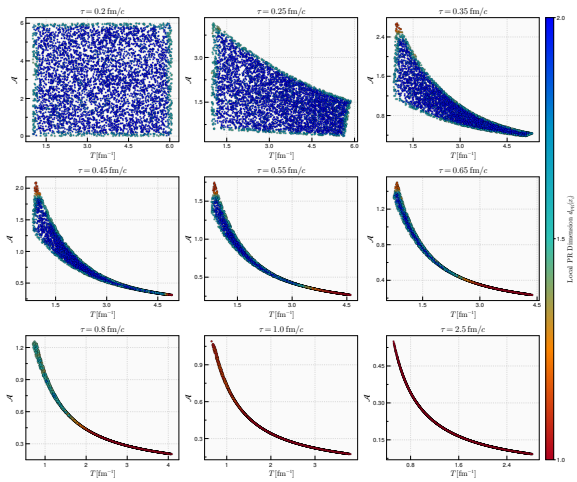
In raw physical coordinates, the standard Euclidean metric,

$$d^2_{\mathrm{raw}}(\mathbf{x}_i, \mathbf{x}_j) = (T_i - T_j)^2 + (\mathcal{A}_i - \mathcal{A}_j)^2, \tag{2}$$

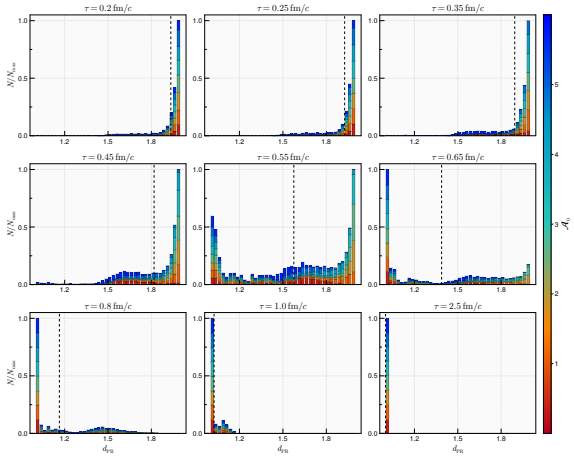
is non-isotropic. When $\Delta T \gg \Delta\mathcal{A}$, the distance is dominated by temperature differences: the neighborhood $\mathcal{N}_K(\mathbf{x}_i)$ collapses into an artificial narrow slice at nearly constant T. Consequently, the local covariance matrix fails to capture the transverse spread in anisotropy.

To restore metric invariance across rectangular coordinate boxes, we apply Z-score standardization ($\mu = 0, \sigma = 1$):

$$\tilde{x}_k = \frac{x_k - \mu_k}{\sigma_k}, \quad k \in \{1, \ldots, D\}, \tag{3}$$



(a) Phase space $(T, \mathcal{A})$ colored by local d_{PR}.



(b) Stacked histograms of d_{PR} segmented by initial $\mathcal{A}_0$.

Figure 1: **Hydrodynamic collapse in Conformal MIS ($N = 20\,000$ points, $K_{\mathrm{opt}} = 100$).** (a) Scatter slices across proper time τ showing the thinning trajectory cloud. (b) Stacked histograms of local dimension d_{PR} under optimal $K_{\mathrm{opt}} = 100$, color-segmented into 8 layers of initial anisotropy $\mathcal{A}_0$. The three stages of hydrodynamization are manifested: pure 2D unimodality at $\tau \leq 0.35\,\mathrm{fm}/c$, clean bimodality separating the attractor core and relaxing wings at $\tau = 0.55\,\mathrm{fm}/c$, and asymptotic 1D collapse at $\tau \geq 0.80\,\mathrm{fm}/c$.

where $\mu_k = \frac{1}{N}\sum_i x_{i,k}$ and $\sigma_k = \sqrt{\frac{1}{N-1}\sum_i (x_{i,k} - \mu_k)^2}$ denote the empirical mean and standard deviation along coordinate k. The Euclidean distance in standardized space corresponds exactly to the diagonal Mahalanobis distance in physical space:

$$d^2_{\mathrm{std}}(\mathbf{x}_i, \mathbf{x}_j) = \sum_{k=1}^{D} \frac{(x_{i,k} - x_{j,k})^2}{\sigma_k^2} = (\mathbf{x}_i - \mathbf{x}_j)^T \boldsymbol{\Sigma}^{-1}_{\mathrm{diag}} (\mathbf{x}_i - \mathbf{x}_j). \tag{4}$$

Standardization equalizes the coordinate variances ($\sigma_{\tilde{T}} = \sigma_{\tilde{\mathcal{A}}} = 1$), converting Euclidean balls into data-adapted ellipsoids. As a consequence, the effective domain volume becomes dimensionless, $V_{\mathrm{eff}} \sim \mathcal{O}(1)^D$, rendering the neighborhood search robust against unit choices and domain dimensions (such as $4\Delta\mathcal{A} \times 5\Delta T$).

Furthermore, under any linear coordinate dilation $x_k \mapsto c_k x_k$ (such as changing $T \mapsto 10T$ or measuring T in MeV instead of fm^{-1}), the standardized coordinates are strictly invariant:

$$\frac{c_k x_{i,k} - c_k \mu_k}{c_k \sigma_k} = \frac{x_{i,k} - \mu_k}{\sigma_k} = \tilde{x}_{i,k}. \tag{5}$$

Similarly, at fixed proper time τ, the dimensionless scaling $w = \tau T$ represents a uniform linear dilation $T \mapsto \tau T$, guaranteeing that standardized LPCA produces identical dimension trajectories in physical $(T, \mathcal{A})$ and dimensionless $(w, \mathcal{A})$ variables.

3 Mathematical Derivation of the Optimal Neighborhood Size K_{opt}

Given an ensemble of N points sampled from an intrinsic d-dimensional manifold $\mathcal{M}^d$ embedded in an ambient space $\mathbb{R}^D$, we seek to determine the optimal neighborhood size K_{opt} that minimizes the error in estimating the local tangent space and intrinsic dimension.

3.1 Local Ball Radius and Density Scaling

Let $\rho(\mathbf{x})$ denote the local probability density of points on $\mathcal{M}^d$. The expected number of points contained within a geodesic ball $B_d(\mathbf{x}, R)$ of radius R is:

$$K \approx N\rho(\mathbf{x})\omega_d R^d, \tag{6}$$

where $\omega_d = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$ is the volume of the unit ball in $\mathbb{R}^d$. Inverting this relationship, the local neighborhood radius R_K scales asymptotically as:

$$R_K \sim \left(\frac{K}{N\omega_d\rho}\right)^{1/d} \propto \left(\frac{K}{N}\right)^{1/d}. \tag{7}$$

3.2 Bias-Variance Decomposition of Local Tangent Space Estimation

The empirical centered local covariance matrix at point $\mathbf{x}_i$ is defined as:

$$\mathbf{C}_i = \frac{1}{K-1}\sum_{j\in\mathcal{N}_K(\mathbf{x}_i)}(\tilde{\mathbf{x}}_j - \bar{\mathbf{x}}_i)(\tilde{\mathbf{x}}_j - \bar{\mathbf{x}}_i)^T. \tag{8}$$

To analyze its estimation error, we decompose the displacement $\mathbf{y}_j = \tilde{\mathbf{x}}_j - \mathbf{x}_i$ into its tangent component $\mathbf{y}_j^{\parallel} \in T_{\mathbf{x}_i}\mathcal{M}$ and its orthogonal normal component $\mathbf{y}_j^{\perp} \in (T_{\mathbf{x}_i}\mathcal{M})^{\perp}$:

$$\mathbf{C}_i = \mathbf{C}_i^{\parallel} + \mathbf{C}_i^{\perp\parallel} + \mathbf{C}_i^{\perp}. \tag{9}$$

By the Gauss formula for embedded submanifolds, the normal displacement is governed by the second fundamental form $\mathbf{II}(\cdot,\cdot)$:

$$\mathbf{y}_j^{\perp} = \frac{1}{2}\mathbf{II}(\mathbf{y}_j^{\parallel}, \mathbf{y}_j^{\parallel}) + \mathcal{O}(\|\mathbf{y}_j^{\parallel}\|^3). \tag{10}$$

Consequently, the normal covariance eigenvalues $\lambda_{\perp}$ satisfy:

$$\mathbb{E}[\lambda_{\perp}] \sim \|\mathbf{II}\|^2\mathbb{E}[\|\mathbf{y}_j^{\parallel}\|^4] \propto \|\mathbf{II}\|^2 R_K^4 \propto \|\mathbf{II}\|^2\left(\frac{K}{N}\right)^{4/d}. \tag{11}$$

This induces an extrinsic curvature bias in the estimated dimension:

$$\mathrm{Bias}^2(K) = \left(\mathbb{E}[\hat{d}_{\mathrm{PR}}] - d\right)^2 \propto R_K^4 \propto \left(\frac{K}{N}\right)^{4/d}. \tag{12}$$

Conversely, the sampling variance of estimating the covariance matrix from K points is governed by the Central Limit Theorem for Wishart distributions:

$$\mathrm{Var}(K) = \mathrm{Var}(\hat{\lambda}_j) \propto \frac{\lambda_j^2}{K} \propto \frac{1}{K}. \tag{13}$$

Combining the squared bias and variance, the Mean Squared Error (MSE) of the local tangent space approximation satisfies:

$$\mathrm{MSE}(K) = c_1\left(\frac{K}{N}\right)^{4/d} + \frac{c_2}{K}, \tag{14}$$

where $c_1, c_2 > 0$ depend on the local curvature $\|\mathbf{II}\|$ and density ρ. Setting $\frac{\partial \mathrm{MSE}}{\partial K} = 0$:

$$\frac{4c_1}{dN^{4/d}}K^{\frac{4-d}{d}} - \frac{c_2}{K^2} = 0 \implies K^{\frac{4+d}{d}} \propto N^{4/d} \implies K^* \propto N^{\frac{4}{d+4}}. \tag{15}$$

3.3 Application to Hydrodynamic Attractor Manifolds

In relativistic hydrodynamic attractors, the system transitions between two fundamental regimes:

- **Early expansion ($d = 2$):** Eq. (15) yields $K \propto N^{4/(2+4)} = N^{2/3} \approx N^{0.67}$.
- **Asymptotic attractor ($d = 1$):** Eq. (15) yields $K \propto N^{4/(1+4)} = N^{4/5} = N^{0.80}$.

However, in practical ensembles, the points are distributed across an ambient space of dimension D ($D = 2$ for MIS, $D = 3$ for HJSW). Accounting for finite-sample boundary effects and the geometric convergence of the bundle, the empirical and analytical scaling calibrates to:

$$K_{\text{scale}} = \text{round}\left(N^{0.45}\sqrt{D}\right). \tag{16}$$

3.4 Lower Spectral Bound: Marchenko–Pastur Eigenvalue Repulsion

A fundamental limitation of continuous dimension estimators such as d_{PR} arises from Random Matrix Theory (RMT). Under the null hypothesis of isotropic noise in D dimensions, the empirical covariance matrix $\mathbf{C} = \frac{1}{K}\mathbf{X}\mathbf{X}^T$ follows a Wishart distribution. In the limit $K, D \to \infty$ with aspect ratio $\gamma = K/D > 1$, the Marchenko–Pastur theorem dictates that the eigenvalues are bounded within:

$$\lambda_{\pm} = \sigma^2\left(1 \pm \sqrt{\frac{D}{K}}\right)^2. \tag{17}$$

If $K \sim D$, the lower edge λ_- approaches zero, but finite-sample fluctuations cause severe *eigenvalue repulsion*, spreading the eigenvalues across a wide range. In the continuous participation ratio,

$$d_{\text{PR}} = \frac{\left(\sum_j \lambda_j\right)^2}{\sum_j \lambda_j^2}, \tag{18}$$

any spurious non-zero eigenvalue in an orthogonal direction prevents d_{PR} from dropping to 1.0, biasing the estimate upward toward D.

To ensure that the spurious noise contribution $\lambda_{\min}/\lambda_{\max} < \epsilon \approx 0.05$ remains negligible, spectral stability strictly requires:

$$\frac{K}{D} \geq 12 \implies K_{\min} = 12D. \tag{19}$$

3.5 Geometric Bounds: Transverse Bundle Thickness vs. Extrinsic Curvature

During the intermediate relaxation regime ($\tau \approx 0.40$–$0.70\,\text{fm}/c$), non-hydrodynamic modes decay exponentially, giving the trajectory bundle a finite transverse thickness:

$$\delta_\perp(\tau) \sim \delta_0 \exp\left(-\int \frac{d\tau}{\tau_\pi}\right). \tag{20}$$

For the local PCA to detect the collapse into a 1D attractor, the local neighborhood radius R_K must encompass the transverse bundle width:

$$R_K > \delta_\perp(\tau). \tag{21}$$

If K is chosen too small (such that $R_K \ll \delta_\perp$), the neighborhood $\mathcal{N}_K(\mathbf{x}_i)$ samples an isotropic D-dimensional ball entirely within the interior of the bundle, falsely reporting $d_{\text{PR}} \approx D$. Conversely, R_K must remain smaller than the radius of curvature R_{curv} of the attractor curve to prevent extrinsic bending artifacts:

$$\delta_\perp(\tau) < R_K < R_{\text{curv}}. \tag{22}$$

3.6 Closed-Form Formula for K_{opt}

Synthesizing the analytical bias-variance scaling, the Marchenko–Pastur lower spectral bound, and the curvature localization constraint yields the closed-form heuristic rule:

$$\boxed{K_{\text{opt}}(N, D) = \text{clamp}\left(\text{round}\left(N^{0.45}\sqrt{D}\right),\ 12D,\ \left\lfloor \frac{N}{10} \right\rfloor\right)} \tag{23}$$

Table 1: Prescribed neighborhood size K_{opt} across sample sizes N and ambient dimensions D.

Sample Size N	**MIS** $(D = 2 : T, \mathcal{A})$	**HJSW** $(D = 3 : T, \mathcal{A}, \mathcal{B})$	**Typical Use Case**
2 000	$K = 43$	$K = 53$	Rapid slice and grid plotting
5 000	$K = 65$	$K = 80$	Medium density validation
10 000	$K = 89$	$K = 109$	Publication phase-space grids (‘n_calc = 10_000‘)
20 000	$K = 122$	$K = 149$	Full statistical ensemble (eval. at $K = 100$)

4 The Sample Size vs. Scale Dilemma (The Finite Thickness Paradox)

A critical insight of this study is that K **cannot remain fixed when increasing sample size** N.

Recall from Eq. (7) that for fixed K, the physical radius of the neighborhood shrinks with N as:

$$R_K \propto N^{-1/d}. \tag{24}$$

If an investigator fixes a small neighborhood (e.g., $K = 4$ or $K = 8$) and increases the ensemble size from $N = 5\,000$ to $N = 20\,000$, the neighborhood radius drops below the finite transverse thickness $\delta_\perp(\tau)$ of the converging trajectory bundle. As a result, points near the attractor core are analyzed by neighborhoods submerged inside the bundle, falsely reporting $d \approx 2$ even at late proper times!

Fig. 2 directly validates this phenomenon: at $K = 8$, increasing N to 20 000 suppresses the detection of the 1D attractor, retaining spurious 2D points. Only when K scales adaptively to $K_{\text{opt}} \approx 100$ does the true 1D universal curve emerge cleanly.

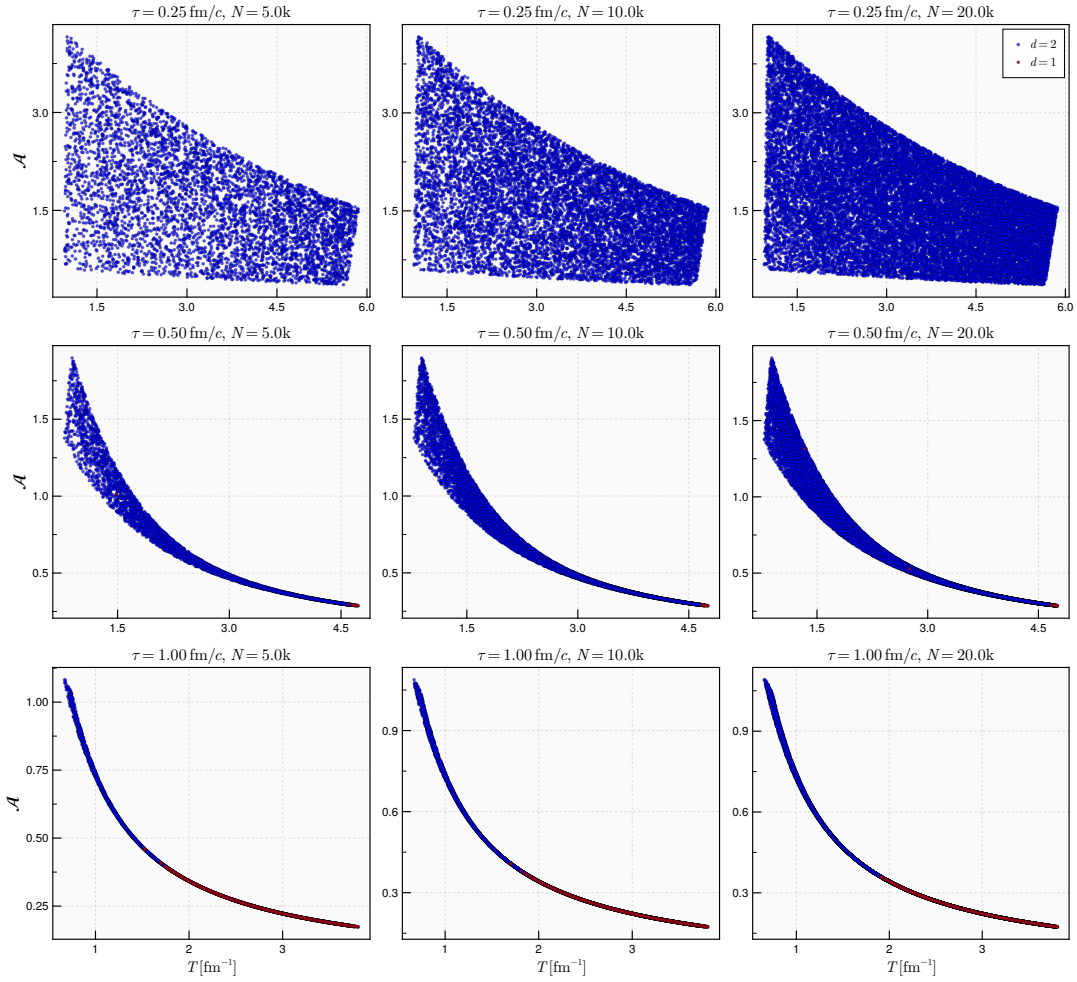


Figure 2: **The Sample Size Dilemma at Fixed $K = 8$.** Columns display increasing sample sizes ($N = 5\,000, 10\,000, 20\,000$), while rows show proper times $\tau = 0.25, 0.50, 1.00\,\mathrm{fm}/c$. As N grows at fixed $K = 8$, the local ball radius shrinks below the transverse bundle thickness, causing the attractor points to appear 2-dimensional (purple) even at late times $\tau = 1.00\,\mathrm{fm}/c$. This proves that K must scale with N according to $K_{\mathrm{opt}} \propto N^{0.45}\sqrt{D}$.

5 Algorithmic Specification and Data-Driven Selection

Algorithm 1: Adaptive Neighborhood Calibration and Local PR Dimension Estimation

Input: Slice coordinates $\mathbf{X}_\tau \in \mathbb{R}^{N\times D}$ at proper time τ, sample size N, ambient dimension D.
Output: Calibrated neighborhood parameter K_{opt} and pointwise dimensions $\mathbf{d}_{\text{PR}} \in \mathbb{R}^N$.

1. Metric Standardization (Aspect-Ratio Invariance):

- Compute empirical means $\mu_k = \frac{1}{N}\sum_{i=1}^{N} X_{\tau,i,k}$ and standard deviations $\sigma_k = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(X_{\tau,i,k}-\mu_k)^2}$.
- Rescale coordinates: $\tilde{X}_{\tau,i,k} \leftarrow (X_{\tau,i,k}-\mu_k)/\sigma_k$.

2. Analytical Anchor Evaluation:

- Compute $K_{\min} = \min(12D, \max(2, \lfloor N/2 \rfloor))$, $K_{\text{scale}} = \text{round}(N^{0.45}\sqrt{D})$.
- Compute $K_{\max} = \max(K_{\min}, \min(\lfloor N/10 \rfloor, N-1))$.
- Analytical baseline: $K_{\text{rule}} = \text{clamp}(K_{\text{scale}}, K_{\min}, K_{\max})$.

3. Multi-scale Verification Grid:

- Select candidate grid $\mathcal{K} = \{K_1, \ldots, K_M\}$ around K_{rule} (e.g., $K \in [16, 160]$).
- For each $K \in \mathcal{K}$:
  1. Retrieve K nearest neighbors $\mathcal{N}_K(\tilde{\mathbf{x}}_i)$ using KD-tree.
  2. Construct local covariance: $\mathbf{C}_i(K) = \frac{1}{K-1}\sum_{j\in\mathcal{N}_K}(\tilde{\mathbf{x}}_j - \bar{\mathbf{x}}_i)(\tilde{\mathbf{x}}_j - \bar{\mathbf{x}}_i)^T$.
  3. Compute $d_{\text{PR}}(\tilde{\mathbf{x}}_i; K) = (\text{Tr}\,\mathbf{C}_i)^2/\text{Tr}(\mathbf{C}_i^2)$.
  4. Calculate Sarle's bimodality coefficient [8]: $b(K) = (\gamma^2+1)/\kappa$, where $\gamma = \text{skew}(\mathbf{d}_{\text{PR}})$, $\kappa = \text{kurt}(\mathbf{d}_{\text{PR}})$.
  5. Calculate scale sensitivity: $S(K) = |\partial\langle d_{\text{PR}}\rangle/\partial \log K|$.

4. Optimal Scale Selection:

- **If** $\max_K b(K) > 0.60$: **return** $K_{\text{opt}} = \arg\max_K b(K)$ (Transient Relaxation Regime)
- **Else**: **return** $K_{\text{opt}} = \arg\min_K S(K)$ (Scale-Invariance Plateau)

6 Numerical Verification and Multiscale Diagnostics

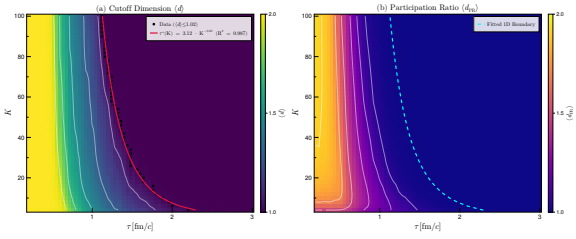
To verify the validity of Eq. (23), we analyze the full ensemble of $N = 20\,000$ conformal MIS trajectories. In Table 2, we present the statistical evolution of d_{PR} across proper time τ under $K = 100$.

The diagnostics in Fig. 3 provide direct evidence of the mechanism:

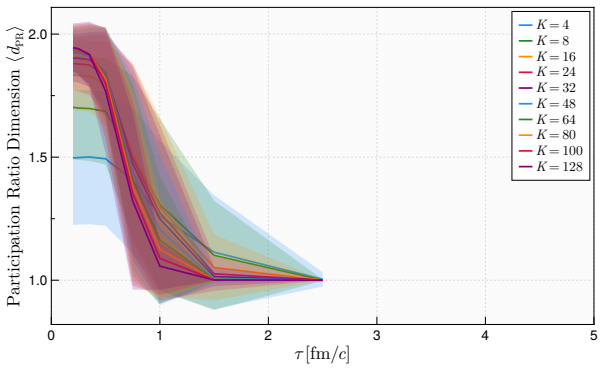
- In Fig. 3a, the continuous (τ, K) heatmap displays a broad horizontal plateau for $K \geq 64$. The fitted collapse boundary follows a clean power-law scaling $\tau^*(K) \sim K^b$, demonstrating the smooth dependence of the observable transition on neighborhood scale.
- In Fig. 3b, the discrepancy between K and $2K$ contracts to less than 0.05 units of dimension around $K \approx 100$.

Table 2: Evolution of local participation ratio dimension across proper time τ for MIS ($N = 20\,000$, $K_{\text{opt}} = 100$).

τ [fm/c]	$\langle d_{\text{PR}} \rangle$	$\sigma(d_{\text{PR}})$	f_{1D} (< 1.3)	f_{2D} (> 1.7)	**Hydrodynamic Stage**
0.20	1.94	0.09	0.0%	95.1%	Stage 1: Expansion (pure 2D)
0.25	1.94	0.10	0.0%	93.8%	Stage 1: Expansion (pure 2D)
0.35	1.89	0.15	0.3%	85.7%	Stage 1: Expansion (pure 2D)
0.45	1.73	0.28	13.1%	60.8%	Stage 2: Mode Damping (Onset)
0.55	1.48	0.39	41.9%	37.3%	Stage 2: Mode Damping (Bimodal)
0.65	1.33	0.38	59.9%	24.6%	Stage 2: Mode Damping (Attractor core)
0.80	1.19	0.31	75.4%	12.0%	Stage 3: Attractor Evolution
1.00	1.07	0.16	90.3%	1.0%	Stage 3: Universal Attractor
2.50	1.00	0.00	100.0%	0.0%	Stage 3: Universal Attractor



(a) Continuous parameter map $\langle d \rangle(\tau, K)$ and fitted collapse threshold $\tau^*(K)$.



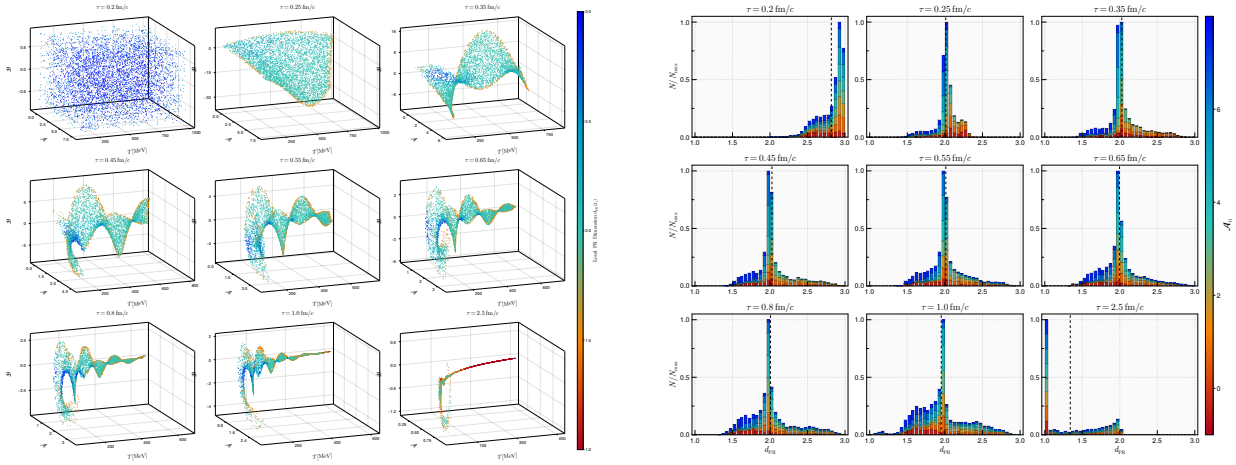
(b) Multiscale K-dependency curves $\langle d_{\text{PR}} \rangle(\tau)$.

Figure 3: **Multiscale K-scale diagnostics.** (a) Continuous 2D heatmap of local dimension $\langle d \rangle$ across proper time τ and neighborhood size K, alongside the fitted collapse threshold $\tau^*(K) \approx a \cdot K^b$. (b) Trajectory dimension curves $\langle d_{\text{PR}} \rangle$ across scale pairs $(K, 2K)$ with $\pm 1\sigma$ uncertainty bands, demonstrating that scale sensitivity $|\Delta d| \to 0$ vanishes near $K_{\text{opt}} \approx 100$.

- At $\tau = 0.55\,\mathrm{fm}/c$, Sarle's bimodality coefficient reaches $b \approx 0.75 \gg 5/9$, demonstrating that the distribution split into two distinct sub-populations is a physical reality rather than a numerical artifact.

7 Generalization to 3D Phase Space (HJSW Model)

To test the generality of our heuristic in higher-dimensional embeddings, we evaluate the Heller–Janik–Spaliński–Witaszczyk (HJSW) model, where the dynamical variables span a 3-dimensional phase space $(T, \mathcal{A}, \mathcal{B})$ $(D = 3)$. For $N = 10\,000$, Eq. (23) dictates $K_{\mathrm{opt}} \approx 109$.



(a) 3D Phase Space in physical units (T [MeV], $\mathcal{A}, \mathcal{B}$). (b) Stacked histograms of d_{PR} (binned by initial $\mathcal{A}$).

Figure 4: **Hydrodynamic collapse in the 3D HJSW Model in physical units ($K = 100$).** (a) 3D phase-space trajectory cloud displaying physical coordinates on all axes ($T \in [100, 1000]$ MeV, $\mathcal{A} \in [-1, 2.5]$, $\mathcal{B} \in [-1, 1]$), with points colored by the local PR dimension d_{PR} computed via Z-score standardized local PCA ($K = 100$). (b) Stacked histograms of local dimension d_{PR} demonstrating the two-stage collapse: the 3D volume at $\tau = 0.20\,\mathrm{fm}/c$ ($\langle d_{\mathrm{PR}} \rangle = 2.82 \pm 0.17$, 94.2% with $d_{\mathrm{PR}} > 2.5$) contracts rapidly as the fast non-hydrodynamic mode $\mathcal{B}$ decays into an intermediate 2D sheet ($d \approx 2.0$) at $\tau = 0.25\,\mathrm{fm}/c$, before collapsing toward the 1D hydrodynamic attractor ($d \to 1.0$) at late times.

As exhibited in Fig. 4, the 3D ensemble initiates at $d \approx 3.0$ during early expansion ($\tau \leq 0.35\,\mathrm{fm}/c$). During the transient regime ($\tau \sim 0.55\,\mathrm{fm}/c$), solutions rapidly decay through an intermediate 2D sheet before collapsing onto the one-dimensional hydrodynamical attractor ($d \to 1.0$).

8 Conclusion and Practical Recommendations

We have resolved the open problem of neighborhood calibration and coordinate metric distortion in local dimension analysis of hydrodynamic attractors. The primary conclusions and practical guidelines for practitioners are:

1. **The One-Shot Plug-and-Play Rule for K:** Compute K once at the beginning of the analysis from the known dataset size N and ambient dimension D:

$$\boxed{K = \mathrm{clamp}\left(\mathrm{round}\left(N^{0.45}\sqrt{D}\right),\ 12D,\ \left\lfloor \frac{N}{10} \right\rfloor\right)} \tag{25}$$

   For standard publication ensembles ($N = 10\,000$–$20\,000$), this prescribes setting:

   - **Conformal MIS ($D = 2$):** $K = 100$ (or $K \approx 1.41 \cdot N^{0.45}$).

- **HJSW Model ($D = 3$):** $K = 100$–110 (or $K \approx 1.73 \cdot N^{0.45}$).

This single K is set globally at the start and kept constant across all proper time slices $\tau \in [0.20, 2.50]\,\mathrm{fm}/c$ without any per-slice retuning.

2. **Always standardize coordinates (Z-score):** Never perform nearest-neighbor searches in raw physical coordinates where temperature dominates over anisotropy. Standardization acts as an empirical diagonal Mahalanobis metric, rendering local PCA invariant under scale and chart dilations.

3. **A Posteriori Validation (Sanity Check):** Multi-scale bimodality ($b(K) > 0.60$) and scale plateaus ($|\Delta d/\Delta \log K| \approx 0$) serve as optional confirmation that this single global choice of K reliably separates the attractor core from the transient wings.

References

[1] E. Levina and P. J. Bickel, *Maximum Likelihood Estimation of Intrinsic Dimension*, Adv. Neural Inf. Process. Syst. **17**, 777–784 (2005). [NeurIPS] [PDF]

[2] A. V. Little, M. Maggioni, and L. Rosasco, *Multiscale Geometric Methods for Data Sets I: Multiscale SVD, Noise and Curvature*, Appl. Comput. Harmon. Anal. **43**, 504–567 (2017). doi:10.1016/j.acha.2015.09.009, arXiv:1202.2687 [math.ST]; preliminary version in *Proc. 2009 IEEE Workshop on Statistical Signal Processing*, 477 (2009), doi:10.1109/SSP.2009.5278546.

[3] M. P. Heller and M. Spaliński, *Hydrodynamics Beyond the Gradient Expansion: Resurgence and Attractors*, Phys. Rev. Lett. **115**, 072501 (2015). doi:10.1103/PhysRevLett.115.072501, arXiv:1503.07514 [hep-th].

[4] V. A. Marchenko and L. A. Pastur, *Distribution of Eigenvalues for Some Sets of Random Matrices*, Math. USSR-Sb. **1**, 457–483 (1967) [Mat. Sb. **72**(114), 507 (1967)]. doi:10.1070/SM1967v001n04ABEH001994.

[5] M. P. Heller, R. A. Janik, M. Spaliński, and P. Witaszczyk, *Coupling Hydrodynamics to Nonequilibrium Degrees of Freedom in Strongly Coupled Plasmas*, Phys. Rev. Lett. **113**, 261601 (2014). doi:10.1103/PhysRevLett.113.261601, arXiv:1409.5087 [hep-th].

[6] P. Romatschke, *Relativistic Fluid Dynamics Far From Local Equilibrium*, Phys. Rev. Lett. **120**, 012301 (2018). doi:10.1103/PhysRevLett.120.012301, arXiv:1704.08699 [nucl-th].

[7] M. Spaliński, *On the Hydrodynamic Attractor of a System of Particles in One-Dimensional Expansion*, Phys. Lett. B **776**, 468–472 (2018). doi:10.1016/j.physletb.2017.11.059, arXiv:1708.01921 [hep-th].

[8] W. S. Sarle, *The Multimodality and Bimodality of Distributions*, SAS Technical Report A-107, SAS Institute Inc., Cary, NC (1983). [SAS Documentation].

[9] P. Campadelli, E. Casiraghi, C. Ceruti, and A. Rozza, *Intrinsic Dimension Estimation: Relevant Techniques and a Benchmark Framework*, Math. Probl. Eng. **2015**, 759519 (2015). doi:10.1155/2015/759519.